

3rd

Given

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$W^{(1)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$W^{(2)} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b^{(2)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$W^{(3)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad b^{(3)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

True class: $y = 1$ Learning rate: $\eta = 0.1$

PART A — SOFTMAX + CROSS ENTROPY

(a) Forward Pass

Layer 1

$$z^{(1)} = W^{(1)}x + b^{(1)}$$

Matrix multiplication:

$$W^{(1)}x = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

First row:

$$(1)(1) + (0)(1) = 1 + 0 = 1$$

Second row:

$$(0)(1) + (1)(1) = 0 + 1 = 1$$

$$W^{(1)}x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Add bias:

$$z^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Apply ReLU:

$$a^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Layer 2

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$$

$$W^{(2)}a^{(1)} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

First row:

$$(1)(1) + (-1)(1) = 1 - 1 = 0$$

Second row:

$$(1)(1) + (1)(1) = 1 + 1 = 2$$

$$W^{(2)}a^{(1)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Add bias:

$$z^{(2)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Apply ReLU elementwise:

$$\text{ReLU}(0) = 0, \quad \text{ReLU}(2) = 2$$

$$a^{(2)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Layer 3

$$s = W^{(3)}a^{(2)} + b^{(3)}$$

$$W^{(3)}a^{(2)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

First row:

$$(1)(0) + (0)(2) = 0$$

Second row:

$$(0)(0) + (1)(2) = 2$$

$$s = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Softmax

$$e^0 = 1$$

$$e^2 = 7.389$$

Denominator:

$$1 + 7.389 = 8.389$$

$$q_1 = \frac{1}{8.389} = 0.119$$

$$q_2 = \frac{7.389}{8.389} = 0.881$$

$$q = \begin{bmatrix} 0.119 \\ 0.881 \end{bmatrix}$$

(b) Cross Entropy Loss

$$L_{CE} = -\log(q_1)$$

$$= -\log(0.119)$$

$$= -(-2.13)$$

$$L_{CE} = 2.13$$

(c) Gradient w.r.t Logits

One-hot label:

$$y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\frac{\partial L}{\partial s} = q - y = \begin{bmatrix} 0.119 \\ 0.881 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.881 \\ 0.881 \end{bmatrix}$$

(d) Backpropagation

Layer 3

$$\nabla W^{(3)} = \delta^{(3)} (a^{(2)})^T$$

$$= \begin{bmatrix} -0.881 \\ 0.881 \end{bmatrix} \begin{bmatrix} 0 & 2 \end{bmatrix}$$

Row 1:

$$-0.881(0) = 0, \quad -0.881(2) = -1.762$$

Row 2:

$$0.881(0) = 0, \quad 0.881(2) = 1.762$$

$$\nabla W^{(3)} = \begin{bmatrix} 0 & -1.762 \\ 0 & 1.762 \end{bmatrix}$$

$$\nabla b^{(3)} = \begin{bmatrix} -0.881 \\ 0.881 \end{bmatrix}$$

Layer 2

ReLU derivative:

$$\text{ReLU}'(z^{(2)}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\delta^{(2)} = \begin{bmatrix} -0.881 \\ 0.881 \end{bmatrix} \odot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.881 \end{bmatrix}$$

$$\nabla W^{(2)} = \begin{bmatrix} 0 \\ 0.881 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0.881 & 0.881 \end{bmatrix}$$

$$\nabla b^{(2)} = \begin{bmatrix} 0 \\ 0.881 \end{bmatrix}$$

Layer 1

$$(W^{(2)})^T = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$(W^{(2)})^T \delta^{(2)} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0.881 \end{bmatrix} = \begin{bmatrix} 0.881 \\ 0.881 \end{bmatrix}$$

ReLU derivative at $z^{(1)} = [1, 1]^T$ is 1:

$$\delta^{(1)} = \begin{bmatrix} 0.881 \\ 0.881 \end{bmatrix}$$

$$\nabla W^{(1)} = \begin{bmatrix} 0.881 \\ 0.881 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.881 & 0.881 \\ 0.881 & 0.881 \end{bmatrix}$$

$$\nabla b^{(1)} = \begin{bmatrix} 0.881 \\ 0.881 \end{bmatrix}$$

(e) Gradient Descent Update

Update rule:

$$\theta_{new} = \theta - 0.1 \nabla \theta$$

$$W_{new}^{(1)} = \begin{bmatrix} 0.9119 & -0.0881 \\ -0.0881 & 0.9119 \end{bmatrix}$$

$$b_{new}^{(1)} = \begin{bmatrix} -0.0881 \\ -0.0881 \end{bmatrix}$$

$$W_{new}^{(2)} = \begin{bmatrix} 1 & -1 \\ 0.9119 & 0.9119 \end{bmatrix}$$

$$b_{new}^{(2)} = \begin{bmatrix} 0 \\ -0.0881 \end{bmatrix}$$

$$W_{new}^{(3)} = \begin{bmatrix} 1 & 0.1762 \\ 0 & 0.8238 \end{bmatrix}$$

$$b_{new}^{(3)} = \begin{bmatrix} 0.0881 \\ -0.0881 \end{bmatrix}$$

PART B — HINGE LOSS

(a) Loss

$$L = \max(0, s_2 - s_1 + 1) = \max(0, 2 - 0 + 1) = 3$$

(b) Gradient

$$\frac{\partial L}{\partial s} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

(c) Backpropagation

$$\nabla W^{(3)} = \begin{bmatrix} 0 & -2 \\ 0 & 2 \end{bmatrix}$$

$$\nabla b^{(3)} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\nabla W^{(2)} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\nabla b^{(2)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\nabla W^{(1)} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\nabla b^{(1)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(d) Updated Parameters

$$W_{new}^{(1)} = \begin{bmatrix} 0.9 & -0.1 \\ -0.1 & 0.9 \end{bmatrix}$$

$$b_{new}^{(1)} = \begin{bmatrix} -0.1 \\ -0.1 \end{bmatrix}$$

$$W_{new}^{(2)} = \begin{bmatrix} 1 & -1 \\ 0.9 & 0.9 \end{bmatrix}$$

$$b_{new}^{(2)} = \begin{bmatrix} 0 \\ -0.1 \end{bmatrix}$$

$$W_{new}^{(3)} = \begin{bmatrix} 1 & 0.2 \\ 0 & 0.8 \end{bmatrix}$$

$$b_{new}^{(3)} = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$